

CS 506: Introduction to Quantum Computing

Multi-Qubit Hadamard Transforms, the Deutsch Problem, and the Deutsch–Jozsa Algorithm

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1 Overview

In this lecture, we explore the action of Hadamard gates applied in parallel across multi-qubit systems and investigate the foundational quantum algorithms demonstrating quantum advantage:

1. **Action of the n -fold Hadamard Gate ($H^{\otimes n}$):** We analyze how applying Hadamard gates in parallel transforms general basis states $|x\rangle = |x_1 x_2 \dots x_n\rangle$, deriving the compact inner-product formula:

$$H^{\otimes n} |\vec{x}\rangle = \frac{1}{\sqrt{2^n}} \sum_{\vec{z} \in \{0,1\}^n} (-1)^{\vec{z} \cdot \vec{x}} |\vec{z}\rangle.$$

2. **The Deutsch Problem:** Given a 1-bit boolean black-box function $f : \{0,1\} \rightarrow \{0,1\}$, we determine whether f is constant or balanced using the Deutsch circuit. We show that a single quantum query suffices, whereas any deterministic classical algorithm requires 2 queries.
3. **The Deutsch–Jozsa Algorithm:** Extending the problem to an n -bit boolean function $f : \{0,1\}^n \rightarrow \{0,1\}$, we show that while a deterministic classical algorithm requires $2^{n-1} + 1$ queries in the worst case, the quantum algorithm solves the problem exactly with a single quantum query.

2 Looking at the Hadamard Gate More Closely

2.1 Single-Qubit Hadamard Gate

Recall the single-qubit Hadamard gate H , whose matrix representation in the computational basis $\{|0\rangle, |1\rangle\}$ is

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

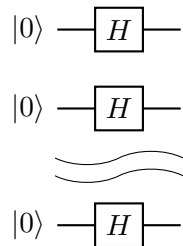
Its action on the standard computational basis states is:

$$H |0\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle), \tag{1}$$

$$H |1\rangle = \frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle). \tag{2}$$

2.2 Action of $H^{\otimes n}$ on the All-Zero State $|0^n\rangle$

Consider an n -qubit register initialized to the all-zero state $|0^n\rangle = |0\rangle^{\otimes n}$, where each qubit passes through an independent Hadamard gate:



The joint operation acting on the n qubits is the tensor product of n Hadamard operators:

$$H^{\otimes n} = \underbrace{H \otimes H \otimes \cdots \otimes H}_{n \text{ times}}.$$

Case $n = 1$:

$$H |0\rangle = \frac{1}{\sqrt{2^1}}(|0\rangle + |1\rangle).$$

Case $n = 2$:

$$\begin{aligned} H^{\otimes 2} |00\rangle &= (H |0\rangle) \otimes (H |0\rangle) \\ &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\ &= \frac{1}{\sqrt{2^2}}(|00\rangle + |01\rangle + |10\rangle + |11\rangle). \end{aligned}$$

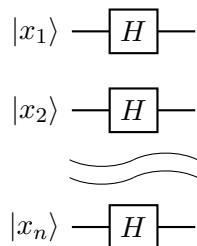
General n : By expanding the tensor product across all n qubits, every combination of n -bit strings appears with identical positive amplitude:

$$H^{\otimes n} |0^n\rangle = \underbrace{H |0\rangle \otimes H |0\rangle \otimes \cdots \otimes H |0\rangle}_{n \text{ times}} = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle. \quad (3)$$

This creates an equal superposition of all 2^n basis states.

2.3 Action of $H^{\otimes n}$ on an Arbitrary Basis State $|x\rangle$

Now consider what happens when the input register is an arbitrary computational basis state $|x\rangle = |x_1 x_2 \dots x_n\rangle$, where each $x_i \in \{0, 1\}$:



If we directly evaluate a product state such as $H |0\rangle \otimes H |1\rangle \otimes H |1\rangle \otimes H |0\rangle$:

$$\left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) \left(\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \right) \left(\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \right) \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) = ?$$

Expanding this expression by brute force becomes excessively cumbersome because one must track the individual signs of all 2^n resulting terms. To simplify this, we take a closer and more structured look at the single-qubit Hadamard transformation.

Notice that for a single bit $x \in \{0, 1\}$:

$$\begin{aligned} H|0\rangle &= \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = \frac{1}{\sqrt{2}}|0\rangle + (-1)^0 \frac{1}{\sqrt{2}}|1\rangle, \\ H|1\rangle &= \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle = \frac{1}{\sqrt{2}}|0\rangle + (-1)^1 \frac{1}{\sqrt{2}}|1\rangle. \end{aligned}$$

Hence, for any $x \in \{0, 1\}$:

$$H|x\rangle = \frac{1}{\sqrt{2}}|0\rangle + (-1)^x \frac{1}{\sqrt{2}}|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^x |1\rangle).$$

We can write this in a unified summation indexing the output state by $z \in \{0, 1\}$:

$$\frac{1}{\sqrt{2}}(|0\rangle + (-1)^x |1\rangle) = \frac{1}{\sqrt{2}}((-1)^{0 \cdot x} |0\rangle + (-1)^{1 \cdot x} |1\rangle) = \frac{1}{\sqrt{2}} \sum_{z \in \{0, 1\}} (-1)^{z \cdot x} |z\rangle.$$

2.4 Multi-Qubit Generalization

Let $\vec{x} = (x_1, x_2, \dots, x_n) \in \{0, 1\}^n$ and $\vec{z} = (z_1, z_2, \dots, z_n) \in \{0, 1\}^n$. Applying $H^{\otimes n}$ to $|\vec{x}\rangle = |x_1\rangle |x_2\rangle \cdots |x_n\rangle$ yields:

$$\begin{aligned} H^{\otimes n} |\vec{x}\rangle &= H|x_1\rangle \otimes H|x_2\rangle \otimes \cdots \otimes H|x_n\rangle \\ &= \bigotimes_{j=1}^n \left(\frac{1}{\sqrt{2}} \sum_{z_j \in \{0, 1\}} (-1)^{z_j x_j} |z_j\rangle \right) \\ &= \frac{1}{\sqrt{2^n}} \sum_{z_1, \dots, z_n \in \{0, 1\}} (-1)^{z_1 x_1 + \cdots + z_n x_n} |z_1 \dots z_n\rangle. \end{aligned}$$

Recognizing the exponent as the standard bitwise inner product modulo 2:

$$\vec{z} \cdot \vec{x} = \sum_{i=1}^n z_i x_i \pmod{2},$$

we obtain the general expression:

$$H^{\otimes n} |\vec{x}\rangle = \frac{1}{\sqrt{2^n}} \sum_{\vec{z} \in \{0, 1\}^n} (-1)^{\vec{z} \cdot \vec{x}} |\vec{z}\rangle \quad (4)$$

3 The Deutsch and Deutsch–Jozsa Problems

3.1 Problem Formulations

- **Deutsch Problem:** We are given a black-box boolean function $f : \{0, 1\} \rightarrow \{0, 1\}$. In a query, we provide an input and observe the output. We are guaranteed that f is either *constant* ($f(0) = f(1)$) or *balanced* ($f(0) \neq f(1)$).

- **Deutsch–Jozsa Problem:** We are given a black-box function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ on n bits. We are guaranteed that f is either:
 - **Constant:** $f(x)$ takes the exact same value for all $x \in \{0, 1\}^n$ (i.e., all 0s or all 1s).
 - **Balanced:** $f(x) = 0$ for exactly half of the inputs (2^{n-1} inputs) and $f(x) = 1$ for the remaining half (2^{n-1} inputs).

Goal: Determine whether f is constant or balanced using the minimum number of oracle queries.

3.2 Classical vs. Quantum Query Complexity

- **Classical Complexity (Deutsch Problem, $n = 1$):** A deterministic classical algorithm requires **2 queries** ($f(0)$ and $f(1)$) to determine whether $f(0) = f(1)$. Knowing only one output yields zero information about whether f is constant or balanced.
- **Classical Complexity (Deutsch–Jozsa Problem, general n):** In the worst case, a deterministic classical algorithm could observe the same function output for 2^{n-1} distinct inputs and still not know if the function is constant or balanced. Therefore, it requires:

$$2^{n-1} + 1 \text{ queries.}$$

- **Quantum Complexity:** In both problems, **only 1 quantum query** is required!

We use the mathematical properties derived for $H^{\otimes n}$ to demonstrate how this quantum speedup works.

4 Quantum Circuits and the Oracle

4.1 The Unitary Oracle U_f

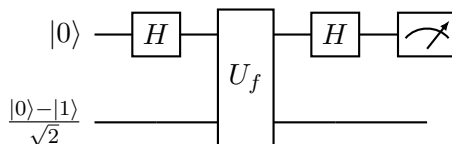
In quantum computation, classical functions (which may not be reversible or one-to-one) are represented by unitary operators using an auxiliary target register. For $x \in \{0, 1\}^n$ and $y \in \{0, 1\}$, the oracle operator U_f is defined as:

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle, \quad (5)$$

where $\oplus$ denotes addition modulo 2 (bitwise XOR).

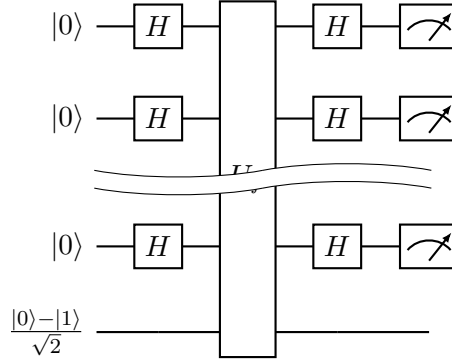
4.2 Deutsch Circuit ($n = 1$)

For the single-qubit Deutsch problem ($n = 1$), the circuit is:



4.3 Deutsch–Jozsa Circuit (General n)

For n input qubits and 1 target qubit, the circuit is:



5 Detailed Mathematical Step-by-Step Analysis

We trace the state of the $(n + 1)$ -qubit system across the time slices $t = 0, 1, 2, 3$ labeled in the Deutsch–Jozsa circuit.

5.1 Time $t = 0$: State Preparation

The n -qubit query register is prepared in $|0^n\rangle$, while the target qubit is prepared in the state $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$:

$$|\psi_0\rangle = |0^n\rangle \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right). \quad (6)$$

5.2 Time $t = 1$: Superposition of Inputs

We apply $H^{\otimes n}$ across the first n qubits while leaving the target qubit untouched:

$$|\psi_1\rangle = (H^{\otimes n} |0^n\rangle) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = \left(\frac{1}{\sqrt{2^n}} \sum_{\vec{x} \in \{0,1\}^n} |\vec{x}\rangle \right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right). \quad (7)$$

Note that because the input state is $|0^n\rangle$, the phase factor $(-1)^{\vec{z} \cdot \vec{0}} = (-1)^0 = 1$, so there are no alternating signs here.

5.3 Time $t = 2$: Oracle Query and Phase Kickback

Next, the unitary oracle U_f is applied to the joint system. By linearity:

$$|\psi_2\rangle = U_f |\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{\vec{x} \in \{0,1\}^n} U_f \left(|\vec{x}\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right).$$

Now consider the action of U_f on each basis component:

$$U_f \left(|\vec{x}\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right) = |\vec{x}\rangle \left(\frac{|0 \oplus f(\vec{x})\rangle - |1 \oplus f(\vec{x})\rangle}{\sqrt{2}} \right).$$

- If $f(\vec{x}) = 0$:

$$\frac{|0 \oplus 0\rangle - |1 \oplus 0\rangle}{\sqrt{2}} = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = (-1)^0 \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right).$$

- If $f(\vec{x}) = 1$:

$$\frac{|0 \oplus 1\rangle - |1 \oplus 1\rangle}{\sqrt{2}} = \frac{|1\rangle - |0\rangle}{\sqrt{2}} = - \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = (-1)^1 \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right).$$

In general, for any $f(\vec{x}) \in \{0, 1\}$:

$$U_f \left(|\vec{x}\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right) = (-1)^{f(\vec{x})} |\vec{x}\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right).$$

This crucial effect is called **phase kickback**: the function evaluation $f(\vec{x})$ is written into the global phase of the corresponding input basis state rather than modifying the target qubit.

Substituting this result back into $|\psi_2\rangle$, we have:

$$|\psi_2\rangle = \left(\frac{1}{\sqrt{2^n}} \sum_{\vec{x} \in \{0,1\}^n} (-1)^{f(\vec{x})} |\vec{x}\rangle \right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right). \quad (8)$$

Notice that the target qubit factors out completely and remains in state $|-\rangle$; we do not care about it further and can ignore it for the remainder of the analysis.

5.4 Time $t = 3$: Final Hadamard Transform

We apply $H^{\otimes n}$ to the n -qubit query register:

$$\begin{aligned} |\psi_3\rangle &= \frac{1}{\sqrt{2^n}} \sum_{\vec{x} \in \{0,1\}^n} (-1)^{f(\vec{x})} H^{\otimes n} |\vec{x}\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{\vec{x} \in \{0,1\}^n} (-1)^{f(\vec{x})} \left(\frac{1}{\sqrt{2^n}} \sum_{\vec{z} \in \{0,1\}^n} (-1)^{\vec{z} \cdot \vec{x}} |\vec{z}\rangle \right) \\ &= \frac{1}{2^n} \sum_{\vec{x} \in \{0,1\}^n} \sum_{\vec{z} \in \{0,1\}^n} (-1)^{f(\vec{x}) + \vec{z} \cdot \vec{x}} |\vec{z}\rangle \\ &= \sum_{\vec{z} \in \{0,1\}^n} \left(\frac{1}{2^n} \sum_{\vec{x} \in \{0,1\}^n} (-1)^{f(\vec{x}) + \vec{z} \cdot \vec{x}} \right) |\vec{z}\rangle. \end{aligned}$$

6 Measurement and Deterministic Decision

We measure the n -qubit register in the computational basis, observing a particular bit-string vector $\vec{z}_i \in \{0, 1\}^n$.

Consider specifically the probability of measuring the all-zero state $\vec{z} = |0^n\rangle$:

$$\Pr[\text{observing } |0^n\rangle] = \left| \frac{1}{2^n} \sum_{\vec{x} \in \{0,1\}^n} (-1)^{f(\vec{x}) + \vec{0} \cdot \vec{x}} \right|^2 = \left(\frac{1}{2^n} \sum_{\vec{x} \in \{0,1\}^n} (-1)^{f(\vec{x})} \right)^2. \quad (9)$$

6.1 Case 1: f is Constant

If f is constant, then $f(\vec{x}) = c \in \{0, 1\}$ for all $\vec{x} \in \{0, 1\}^n$:

$$\sum_{\vec{x} \in \{0,1\}^n} (-1)^{f(\vec{x})} = \sum_{\vec{x} \in \{0,1\}^n} (-1)^c = (-1)^c \sum_{\vec{x} \in \{0,1\}^n} 1 = (-1)^c 2^n.$$

Substituting this back into the probability equation:

$$\Pr[\text{observing } |0^n\rangle] = \left(\frac{1}{2^n} \cdot (-1)^c 2^n \right)^2 = ((-1)^c)^2 = 1. \quad (10)$$

Because the total probability across all states must equal 1, the probability of measuring any other bit string $\vec{z} \neq \vec{0}$ is strictly 0:

If f is constant, we observe $|0^n\rangle$ with probability 1 (we cannot observe anything else).

6.2 Case 2: f is Balanced

If f is balanced, then $f(\vec{x}) = 0$ for exactly half of the inputs ($\frac{1}{2}2^n = 2^{n-1}$) and $f(\vec{x}) = 1$ for the remaining half ($\frac{1}{2}2^n = 2^{n-1}$). Therefore:

$$\sum_{\vec{x} \in \{0,1\}^n} (-1)^{f(\vec{x})} = \sum_{\vec{x}: f(\vec{x})=0} (+1) + \sum_{\vec{x}: f(\vec{x})=1} (-1) = 2^{n-1} - 2^{n-1} = 0.$$

Substituting this into the probability expression:

$$\Pr[\text{observing } |0^n\rangle] = \left(\frac{1}{2^n} \cdot 0 \right)^2 = 0. \quad (11)$$

If f is balanced, the probability of observing $|0^n\rangle$ is strictly 0.

Therefore, the register is guaranteed to collapse into some non-zero bit string $\vec{z} \neq \vec{0}$.

6.3 Decision Rule Summary

With a single quantum evaluation of U_f , we measure the query register and apply the following rule:

Measurement Outcome	$\Pr[\text{Outcome} \mid f \text{ is constant}]$	$\Pr[\text{Outcome} \mid f \text{ is balanced}]$
$\vec{z} = 0^n\rangle = 00 \dots 0\rangle$	1	0
$\vec{z} \neq 0^n\rangle$ (any other string)	0	1

Table 1: Outcome probabilities for constant vs. balanced functions in the Deutsch–Jozsa algorithm.

- If the observed bit string is **$|00 \dots 0\rangle$** , the function is **constant**.
- If the observed bit string is **any other state** ($\neq |00 \dots 0\rangle$), the function is **balanced**.

This provides a deterministic, exact solution to the problem with zero error probability using just **1 quantum query**, illustrating a provable exponential separation over deterministic classical algorithms ($2^{n-1} + 1$ queries).